Quiz 5 Preparation Sheet
Multivariable Calculus

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How To Use This Sheet

This sheet is built for a short quiz, so the priority is speed, recognition, and correct setup. The syllabus is: triple integrals in cylindrical coordinates, spherical coordinates, change of variables, vector fields, line integrals, the Fundamental Theorem for Line Integrals, Green's theorem, curl, and divergence.

Use the definitions first, then test yourself with the conceptual questions, then do the extracted problem-set questions. The short answer key is intentionally compact: it is meant for checking, not for replacing the full computations in your solution PDFs.

1 Definitions And Core Facts

1.1 Triple Integrals

A triple integral

$$\iiint_E f(x, y, z)\, dV$$

adds the values of a function over a three-dimensional solid E. If $f = 1$, then

$$\iiint_E 1\, dV$$

is the volume of E.

For rectangular coordinates,

$$dV = dx\, dy\, dz$$

up to order, depending on the chosen order of integration.

1.2 Cylindrical Coordinates

Cylindrical coordinates are

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

The radial coordinate is

$$r = \sqrt{x^2 + y^2}.$$

The volume element is

$$dV = r\, dz\, dr\, d\theta.$$

The extra factor r is the Jacobian.

Use cylindrical coordinates when the problem has cylinders, circular disks, cones of the form $z = \sqrt{x^2 + y^2}$, paraboloids $z = a - br^2$, or expressions involving $x^2 + y^2$.

1.3 Spherical Coordinates

Spherical coordinates are

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

Here ρ is distance from the origin, θ is the angle in the xy-plane, and ϕ is the angle from the positive z-axis.

Important identities:

$$x^2 + y^2 + z^2 = \rho^2, \qquad x^2 + y^2 = \rho^2\sin^2\phi, \qquad z = \rho\cos\phi.$$

The volume element is

$$dV = \rho^2\sin\phi\, d\rho\, d\phi\, d\theta.$$

Use spherical coordinates for spheres, balls, spherical shells, cones through the origin, and integrands involving $x^2 + y^2 + z^2$.

1.4 Change Of Variables In Double Integrals

If

$$x = x(u, v), \qquad y = y(u, v),$$

then

$$\iint_R f(x, y)\, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv.$$

The Jacobian is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}.$$

The main steps are:

1. Rewrite the boundary curves in terms of u, v.
2. Find the new region S.
3. Compute the Jacobian.
4. Replace the integrand and area element.
5. Integrate over the simpler region.

1.5 Vector Fields

A vector field assigns a vector to each point. In two variables,

$$\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

In three variables,

$$\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

Vector fields can represent force, velocity, flow, electric fields, gradients, or direction fields.

1.6 Line Integrals Of Scalar Functions

For a scalar function f along a curve C parametrized by $\mathbf{r}(t)$, $a \leq t \leq b$,

$$\int_C f\, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\|\, dt.$$

This adds the scalar value along the curve, weighted by arc length.

1.7 Line Integrals Of Vector Fields

For a vector field $\mathbf{F}$ along a parametrized curve $\mathbf{r}(t)$,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

In two dimensions, if $\mathbf{F} = \langle P, Q \rangle$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\, dx + Q\, dy.$$

This measures work done by the field along the curve.

1.8 Conservative Vector Fields

A vector field $\mathbf{F}$ is conservative if there exists a scalar potential function ϕ such that

$$\mathbf{F} = \nabla\phi.$$

Then

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

depends only on the endpoints, not the path.

For $\mathbf{F} = \langle P, Q\rangle$ on a simply connected domain, a quick test is

$$P_y = Q_x.$$

If true, the field is conservative. If false, it is not conservative.

For $\mathbf{F} = \langle P, Q, R\rangle$ on a simply connected domain, a quick test is

$$\operatorname{curl}\mathbf{F} = \mathbf{0}.$$

1.9 Fundamental Theorem For Line Integrals

If $\mathbf{F} = \nabla\phi$, and C goes from point A to point B, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A).$$

This is the vector-field analogue of

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

1.10 Green's Theorem

Let C be a positively oriented, simple, closed curve enclosing region D. If

$$\mathbf{F} = \langle P, Q\rangle,$$

then

$$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA.$$

Positive orientation means counterclockwise orientation.

Green's theorem converts a line integral around a closed curve into a double integral over the region inside the curve.

1.11 Curl

For a 3D vector field

$$\mathbf{F} = \langle P, Q, R\rangle,$$

the curl is

$$\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F} = \langle R_y - Q_z,\, P_z - R_x,\, Q_x - P_y\rangle\,.$$

Curl measures local rotation. If $\operatorname{curl}\mathbf{F} = \mathbf{0}$ on a simply connected domain, then $\mathbf{F}$ is conservative.

For a 2D field $\mathbf{F} = \langle P, Q\rangle$, the scalar curl is

$$Q_x - P_y.$$

This is exactly the integrand in Green's theorem.

1.12 Divergence

For

$$\mathbf{F} = \langle P, Q, R \rangle,$$

the divergence is

$$\operatorname{div} \mathbf{F} = P_x + Q_y + R_z.$$

Divergence measures net outward flow from a point. Positive divergence means the field behaves like a source; negative divergence means it behaves like a sink.

2 Vector Calculus Revision Sprint: Line Integrals & Green's Theorem

This section is your complete self-study guide for the vector calculus topics on Quiz 5. It is structured as: **Definitions** → **Formal Theorems** → **Intuition & Diagrams** → **Full Worked Examples from Problem Sets** → **Realistic Quiz Practice**.

2.1 Core Definitions

Parametrized Curves and the Line Element

A **smooth curve** C is given by a parametrization

$$\mathbf{r}(t) = \langle x(t),\, y(t) \rangle \quad (\text{or } \langle x(t), y(t), z(t) \rangle), \qquad a \le t \le b.$$

The **tangent vector** is $\mathbf{r}'(t) = \langle x'(t), y'(t) \rangle$ and the **vector line element** is

$$d\mathbf{r} = \mathbf{r}'(t)\, dt = \langle dx, dy \rangle.$$

The **arc-length element** is $ds = \|\mathbf{r}'(t)\|\, dt$.

Line Integral of a Scalar Function

For a scalar f,

$$\int_C f\, ds = \int_a^b f(\mathbf{r}(t))\, \|\mathbf{r}'(t)\|\, dt.$$

This is **orientation-independent**: it measures total "weight" along the curve.

Line Integral of a Vector Field (Work Integral)

For a vector field $\mathbf{F} = \langle P, Q \rangle$ (or $\langle P, Q, R \rangle$),

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_C P\, dx + Q\, dy.$$

This is **orientation-dependent**: reversing the curve changes the sign.

Physical meaning: If $\mathbf{F}$ is a force field and C is the trajectory of a particle, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is the **work** done by the force.

Conservative Vector Fields and Potential Functions

A vector field $\mathbf{F} = \langle P, Q \rangle$ on a domain D is **conservative** if there exists a scalar function ϕ (called the **potential function** or **scalar potential**) such that

$$\mathbf{F} = \nabla\phi = \left\langle \frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right\rangle.$$

Physical meaning: A conservative field conserves energy. The work done moving a particle depends only on where you start and where you end up, not the path you take.

Test for conservativeness in $\mathbb{R}^2$: If P and Q have continuous partial derivatives on a simply connected domain, then

$$\mathbf{F} \text{ is conservative} \iff P_y = Q_x.$$

How to find ϕ: Integrate $\phi_x = P$ with respect to x, obtaning $\phi = \int P\,dx + g(y)$. Then differentiate with respect to y and set equal to Q to find $g(y)$.

Scalar Curl (2D)

For $\mathbf{F} = \langle P, Q \rangle$, the **scalar curl** (or 2D curl) is

$$\operatorname{curl}_z \mathbf{F} = Q_x - P_y.$$

This is the z-component of $\nabla \times \mathbf{F}$. It measures the tendency of the field to cause local rotation. The field is conservative on a simply connected domain iff this equals zero everywhere.

2.2 The Three Key Theorems (Formal Statements)

Theorem 1: Fundamental Theorem for Line Integrals (FTLI)

Statement. Let C be a smooth curve parametrized by $\mathbf{r}(t)$, $a \le t \le b$, with start point $A = \mathbf{r}(a)$ and end point $B = \mathbf{r}(b)$. Let ϕ be a differentiable function whose gradient $\mathbf{F} = \nabla\phi$ is continuous on C. Then

$$\boxed{\int_C \nabla\phi \cdot d\mathbf{r} = \phi(B) - \phi(A).}$$

Consequences:

- If C is a **closed curve** (so $A = B$) and $\mathbf{F}$ is conservative, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.
- The integral is **path independent**: any smooth curve from A to B gives the same value.

Analogy with single-variable calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a) \quad \longleftrightarrow \quad \int_C \nabla\phi \cdot d\mathbf{r} = \phi(B) - \phi(A).$$

The FTLI tells us that antiderivatives in the multivariable world are potential functions.

Theorem 2: Green's Theorem

Statement. Let C be a positively oriented (counterclockwise), piecewise-smooth, simple closed curve in the plane, and let D be the region it encloses. If P and Q have continuous partial derivatives on an open set containing D, then

$$\boxed{\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA.}$$

Key points:

- "Simple closed curve" means it does not cross itself.
- "Positively oriented" means the enclosed region D is always on your **left** as you traverse C. Equivalently, the boundary is traversed counterclockwise.
- If the curve is clockwise, the right-hand side gets a **negative sign**.
- The integrand $Q_x - P_y$ is exactly the scalar curl of $\mathbf{F}$.

Very useful corollary — area from a line integral:

$$\text{Area}(D) = \oint_C x\,dy = -\oint_C y\,dx = \tfrac{1}{2}\oint_C x\,dy - y\,dx.$$

This follows from Green's theorem with $(P, Q) = (0, x)$, giving $Q_x - P_y = 1$, so the double integral over D equals $\iint_D 1\,dA = \text{Area}(D)$.

Theorem 3: The Conservative Field Equivalences

On a simply connected domain, the following are equivalent:

1. $\mathbf{F}$ is conservative (i.e., $\mathbf{F} = \nabla\phi$).
2. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C.
3. $\int_C \mathbf{F} \cdot d\mathbf{r}$ is path independent.
4. $P_y = Q_x$ everywhere on the domain.

Warning: Property 4 alone ($P_y = Q_x$ everywhere) is *not* sufficient if the domain is not simply connected (e.g., it has a hole). The classic example is

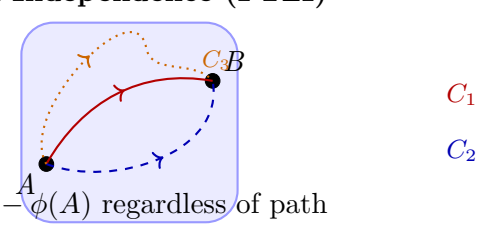
$$\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$$

on $\mathbb{R}^2 \setminus \{(0, 0)\}$: it satisfies $P_y = Q_x$ everywhere it is defined, yet the circulation around any circle enclosing the origin is $2\pi \neq 0$.

2.3 Intuition and Diagrams

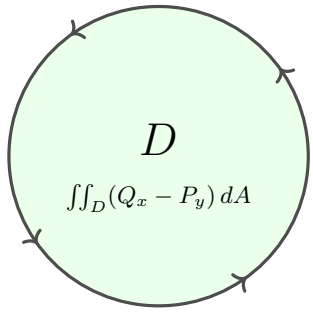
Diagram 1: Path Independence (FTLI)

Path Independence (FTLI)



$\phi(B) - \phi(A)$ regardless of path

Green's Theorem



$$= \oint_C P\, dx + Q\, dy$$

Positive (CCW) orientation

Key intuition for Green's Theorem: Imagine the scalar curl $Q_x - P_y$ as the "density of tiny rotation" at each point inside D. Green's theorem says that if you add up all those tiny rotations over the interior, you get the total circulation around the boundary. Regions where the field spins, clockwise cancel with counterclockwise regions, and only the net boundary circulation survives.

2.4 Fully Worked Problem-Set Examples (Complete Problem Set 10)

The problems below are the complete Problem Set 10 as set by Prof. Rajendra Bhatia. Every solution is written out in full with all intermediate steps.

Problem 1(i) — Path-Independent Integral via FTLI

Problem. Show that the line integral $\int_C 2xe^{-y}\, dx + (2y - x^2e^{-y})\, dy$ is independent of path, and evaluate it for any path C from $(1, 0)$ to $(2, 1)$.

Solution.

Step 1: Identify P, Q, and test for conservativeness.

Set $P(x, y) = 2xe^{-y}$ and $Q(x, y) = 2y - x^2e^{-y}$. Both are defined and have continuous partial derivatives on all of $\mathbb{R}^2$, which is simply connected. Compute:

$$P_y = \frac{\partial}{\partial y}(2xe^{-y}) = -2xe^{-y}, \qquad Q_x = \frac{\partial}{\partial x}(2y - x^2e^{-y}) = -2xe^{-y}.$$

Since $P_y = Q_x = -2xe^{-y}$ everywhere, $\mathbf{F} = \langle P, Q\rangle$ is conservative and the integral is path independent.

Step 2: Find the scalar potential ϕ.

We need $\phi_x = P$ and $\phi_y = Q$. Integrate $\phi_x = 2xe^{-y}$ w.r.t. x:

$$\phi(x, y) = x^2e^{-y} + g(y),$$

where $g(y)$ is unknown. Differentiate w.r.t. y and match with Q:

$$\phi_y = -x^2e^{-y} + g'(y) \overset{!}{=} 2y - x^2e^{-y} \implies g'(y) = 2y \implies g(y) = y^2.$$

So $\phi(x, y) = x^2e^{-y} + y^2$.

Step 3: Apply the FTLI.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(2, 1) - \phi(1, 0) = (4e^{-1} + 1) - (e^0 + 0) = \frac{4}{e} + 1 - 1 = \boxed{\frac{4}{e}}.$$

Problem 1(ii) — Path-Independent Integral via FTLI

Problem. Show that $\int_C \sin y\,dx + (x\cos y - \sin y)\,dy$ is independent of path and evaluate it for any path C from $(2,0)$ to $(1,\pi)$.

Solution.
Step 1: Test for conservativeness.
$P = \sin y$, $Q = x\cos y - \sin y$; both C^∞ on $\mathbb{R}^2$.

$$P_y = \frac{\partial}{\partial y}(\sin y) = \cos y, \qquad Q_x = \frac{\partial}{\partial x}(x\cos y - \sin y) = \cos y.$$

$P_y = Q_x$ everywhere, so $\mathbf{F}$ is conservative and the integral is path independent.
Step 2: Recover ϕ.
Integrate $\phi_x = \sin y$ w.r.t. x: $\phi(x,y) = x\sin y + g(y)$. Differentiate w.r.t. y and match with Q:

$$x\cos y + g'(y) = x\cos y - \sin y \implies g'(y) = -\sin y \implies g(y) = \cos y.$$

Potential: $\phi(x,y) = x\sin y + \cos y$.
Step 3: Evaluate via FTLI.

$$\phi(1,\pi) = 1\cdot\sin\pi + \cos\pi = 0 + (-1) = -1,$$
$$\phi(2,0) = 2\cdot\sin 0 + \cos 0 = 0 + 1 = 1.$$

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(1,\pi) - \phi(2,0) = -1 - 1 = \boxed{-2}.$$

Problem 2 — Winding Number Field on a Punctured Plane

Setup. On $\mathbb{R}^2 \setminus \{(0,0)\}$, consider

$$\mathbf{F}(x,y) = \left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right\rangle = \langle P, Q\rangle.$$

Preliminary: $P_y = Q_x$ but $\mathbf{F}$ is not globally conservative.
By the quotient rule:

$$P_y = \frac{-(x^2+y^2) + y\cdot 2y}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad Q_x = \frac{(x^2+y^2) - x\cdot 2x}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

So $P_y = Q_x$ on the punctured plane. However, the domain has a hole at the origin and is *not* simply connected. We cannot apply the conservativeness criterion. We must integrate directly.

Problem 2(i). C = unit circle, anticlockwise.

Solution.
Parametrize the unit circle: $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$. Then $dx = -\sin t\,dt$, $dy = \cos t\,dt$, and $x^2 + y^2 = 1$.
Substitute P, Q:

$$P\,dx + Q\,dy = \frac{-\sin t}{1}(-\sin t\,dt) + \frac{\cos t}{1}(\cos t\,dt) = (\sin^2 t + \cos^2 t)\,dt = 1\,dt.$$

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} 1\,dt = \boxed{2\pi}.$$

This is nonzero even though $P_y = Q_x$, confirming the field is not conservative on a domain with a hole.

Problem 2(ii). C = circle of radius 2, anticlockwise.

Solution.

Parametrize: $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$. Then $dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$, and $x^2 + y^2 = 4$.

Substitute:

$$P\,dx + Q\,dy = \frac{-2\sin t}{4}(-2\sin t\,dt) + \frac{2\cos t}{4}(2\cos t\,dt) = \frac{4\sin^2 t}{4}\,dt + \frac{4\cos^2 t}{4}\,dt = (\sin^2 t + \cos^2 t)\,dt = 1\,dt.$$

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} 1\,dt = \boxed{2\pi}.$$

Key observation. The integral equals 2π for a circle of *any* radius, as long as the circle encloses the origin. This field is the classical example of a field that is locally irrotational ($P_y = Q_x$) yet globally non-conservative. Its total circulation around the origin equals 2π regardless of the path, which is the topological *winding number* of the curve times 2π.

Problem 3(i) — Green's Theorem over a Triangle

Problem. Evaluate $\oint_C xy^2\,dx + 2x^2y\,dy$, where C is the positively oriented (CCW) boundary of the triangle with vertices $(0,0)$, $(2,2)$, $(2,4)$.

Solution.

Step 1: Compute the scalar curl.

$P = xy^2$, $Q = 2x^2y$.

$$Q_x = \frac{\partial}{\partial x}(2x^2y) = 4xy, \qquad P_y = \frac{\partial}{\partial y}(xy^2) = 2xy. \qquad \Longrightarrow \qquad Q_x - P_y = 4xy - 2xy = 2xy.$$

Step 2: Identify the triangular region D and find the integration bounds.

The three vertices are $O = (0,0)$, $A = (2,2)$, $B = (2,4)$. Find the equations of the three sides:

- Side OA: passes through $(0,0)$ and $(2,2)$, slope $= 1$. Equation: $y = x$, for $0 \le x \le 2$.
- Side AB: passes through $(2,2)$ and $(2,4)$, it is the vertical segment $x = 2$, for $2 \le y \le 4$.
- Side BO: passes through $(2,4)$ and $(0,0)$, slope $= \frac{4}{2} = 2$. Equation: $y = 2x$, for $0 \le x \le 2$.

For $0 \le x \le 2$, the lower boundary of D is $y = x$ and the upper boundary is $y = 2x$. By Green's theorem:

$$\oint_C xy^2\,dx + 2x^2y\,dy = \iint_D 2xy\,dA = \int_0^2 \int_x^{2x} 2xy\,dy\,dx.$$

Step 3: Evaluate the inner integral (in y).

$$\int_x^{2x} 2xy\,dy = 2x\left[\frac{y^2}{2}\right]_{y=x}^{y=2x} = 2x\cdot\frac{(2x)^2 - x^2}{2} = 2x\cdot\frac{4x^2 - x^2}{2} = 2x\cdot\frac{3x^2}{2} = 3x^3.$$

Step 4: Evaluate the outer integral (in x).

$$\int_0^2 3x^3\,dx = \left[\frac{3x^4}{4}\right]_0^2 = \frac{3\cdot 16}{4} = \boxed{12}.$$

Problem 3(ii) — Green's Theorem over a Rectangle

Problem. Evaluate $\oint_C \cos y\,dx + x^2 \sin y\,dy$, where C is the positively oriented boundary of the rectangle with vertices $(0,0)$, $(5,0)$, $(5,2)$, $(0,2)$.

Solution.
Step 1: Compute the scalar curl.
$P = \cos y$, $Q = x^2 \sin y$.

$$P_y = \frac{\partial}{\partial y}(\cos y) = -\sin y, \qquad Q_x = \frac{\partial}{\partial x}(x^2 \sin y) = 2x \sin y.$$

$$Q_x - P_y = 2x \sin y - (-\sin y) = (2x+1)\sin y.$$

Step 2: Set up the double integral over the rectangle $D = [0,5] \times [0,2]$.

$$\iint_D (2x+1)\sin y\,dA = \int_0^5 \int_0^2 (2x+1)\sin y\,dy\,dx.$$

Step 3: The integrand is separable; compute each factor.
Inner integral (in y):

$$\int_0^2 \sin y\,dy = \big[-\cos y\big]_0^2 = -\cos 2 + \cos 0 = 1 - \cos 2.$$

Outer integral (in x):

$$\int_0^5 (2x+1)\,dx = \big[x^2 + x\big]_0^5 = 25 + 5 = 30.$$

Step 4: Multiply.

$$\iint_D (2x+1)\sin y\,dA = 30 \cdot (1 - \cos 2) = \boxed{30(1-\cos 2)}.$$

Problem 3(iii) — Green's Theorem: Area of a Disk

Problem. Evaluate $\oint_C x\,dy - y\,dx$, where C is the circle of radius a, positively oriented (counterclockwise).

Solution.
Step 1: Identify P, Q, and compute the scalar curl.
Write the integral as $\oint_C P\,dx + Q\,dy$ with $P = -y$, $Q = x$.

$$P_y = -1, \qquad Q_x = 1. \qquad \Longrightarrow \qquad Q_x - P_y = 1 - (-1) = 2.$$

Step 2: Apply Green's Theorem.
The region D enclosed by the circle of radius a is a disk with area πa^2:

$$\oint_C x\,dy - y\,dx = \iint_D 2\,dA = 2 \cdot \text{Area}(D) = 2\pi a^2.$$

Answer: $\boxed{2\pi a^2}$.

Key takeaway. This gives the important formula $\text{Area}(D) = \frac{1}{2}\oint_C x\,dy - y\,dx$, which lets you compute the area of any region from just its boundary parametrization.

Independent verification by direct parametrization.

Let $x = a\cos t$, $y = a\sin t$, $0 \le t \le 2\pi$. Then $dy = a\cos t\,dt$, $dx = -a\sin t\,dt$:

$$\begin{aligned}\oint_C x\,dy - y\,dx &= \int_0^{2\pi} \big[a\cos t \cdot a\cos t - a\sin t \cdot (-a\sin t)\big]\,dt \\ &= \int_0^{2\pi} a^2(\cos^2 t + \sin^2 t)\,dt = a^2\int_0^{2\pi} 1\,dt = 2\pi a^2. \quad \checkmark\end{aligned}$$

2.5 Realistic Quiz Practice: Five Likely Questions

A 30-minute quiz with five questions will typically have: one conservative-field identification+potential question, one FTLI evaluation, one straight Green's theorem computation, one orientation or domain reasoning question, and one "should I use Green's or FTLI?" recognition question.

Practice Quiz Question 1. Determine whether $\mathbf{F} = \langle y^2 + 2x,\ 2xy + 3y^2\rangle$ is conservative. If it is, find the potential function $\phi(x, y)$.

Full Solution.

$P = y^2 + 2x$, $Q = 2xy + 3y^2$.

$$P_y = 2y, \qquad Q_x = 2y.$$

Equal, and the domain is all of $\mathbb{R}^2$ (simply connected), so $\mathbf{F}$ is conservative.

Find ϕ: integrate $\phi_x = y^2 + 2x$ with respect to x:

$$\phi = xy^2 + x^2 + g(y).$$

Differentiate: $\phi_y = 2xy + g'(y) = 2xy + 3y^2 \implies g'(y) = 3y^2 \implies g(y) = y^3$.

$$\boxed{\phi(x, y) = xy^2 + x^2 + y^3.}$$

Practice Quiz Question 2. Let $\mathbf{F} = \langle 3x^2y^2,\ 2x^3y\rangle$. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ along any curve from $(-1, 2)$ to $(2, 3)$.

Full Solution.

Check: $P_y = 6x^2y$, $Q_x = 6x^2y$. Conservative.

Find ϕ: $\phi_x = 3x^2y^2 \implies \phi = x^3y^2 + g(y)$. $\phi_y = 2x^3y + g'(y) = 2x^3y \implies g'(y) = 0 \implies g = C$.

So $\phi(x, y) = x^3y^2$.

Apply FTLI:

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(2, 3) - \phi(-1, 2) = (8 \cdot 9) - (-1 \cdot 4) = 72 + 4 = 76.$$

Answer: 76.

Practice Quiz Question 3. Use Green's theorem to evaluate

$$\oint_C (x^2 - y^2)\,dx + 2xy\,dy$$

where C is the boundary of the square $[0, 1] \times [0, 1]$, oriented positively (counterclockwise).

Full Solution.
$P = x^2 - y^2$, $Q = 2xy$.

$$Q_x = 2y, \qquad P_y = -2y.$$

Scalar curl: $Q_x - P_y = 2y - (-2y) = 4y$.

$$\iint_D 4y\,dA = \int_0^1 \int_0^1 4y\,dy\,dx = \int_0^1 \left[2y^2\right]_0^1 dx = \int_0^1 2\,dx = 2.$$

Answer: 2.

Practice Quiz Question 4. Use Green's theorem to find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Full Solution.
Parametrize the ellipse counterclockwise: $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$.
Using the area formula Area $= \frac{1}{2}\oint_C x\,dy - y\,dx$:

$$dx = -a\sin t\,dt, \qquad dy = b\cos t\,dt.$$

$$\text{Area} = \frac{1}{2}\int_0^{2\pi} \Big[(a\cos t)(b\cos t) - (b\sin t)(-a\sin t)\Big]dt = \frac{1}{2}\int_0^{2\pi} ab\big(\cos^2 t + \sin^2 t\big)dt = \frac{1}{2}\int_0^{2\pi} ab\,dt = \pi ab.$$

Answer: πab.

Practice Quiz Question 5. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ where $\mathbf{F} = \langle e^x \sin y,\ e^x \cos y\rangle$ and C is the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$.
Full Solution.
Step 1: Check conservativeness. $P = e^x \sin y$, $Q = e^x \cos y$.

$$P_y = e^x \cos y, \qquad Q_x = e^x \cos y.$$

Equal, so the field is conservative.
Step 2: Find ϕ. Integrate $\phi_x = e^x \sin y$ with respect to x:

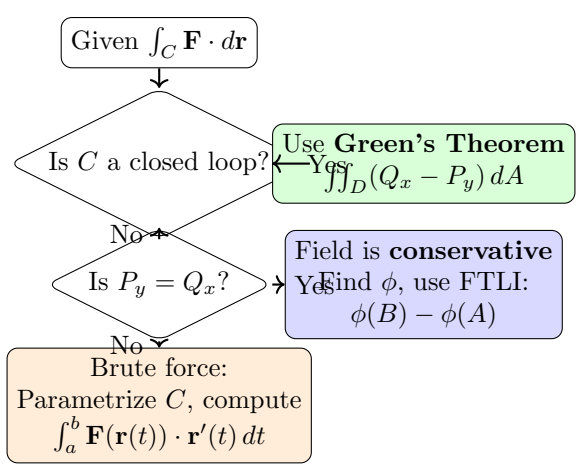
$$\phi = e^x \sin y + g(y).$$

Differentiate: $\phi_y = e^x \cos y + g'(y) = e^x \cos y \implies g'(y) = 0 \implies g = C$.
So $\phi(x,y) = e^x \sin y$.
Step 3: Apply FTLI. (The path does not matter; use start $(0,0)$ and end $(1,1)$.)

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(1,1) - \phi(0,0) = e^1 \sin(1) - e^0 \sin(0) = e\sin 1.$$

Answer: $e\sin 1$.

Summary: The Setup Decision Tree



3 Theoretical Short-Question Quiz

A. Coordinates And Setup

1. What is the difference between r and ρ? **Answer:** $r = \sqrt{x^2 + y^2}$ is distance from the z-axis; $\rho = \sqrt{x^2 + y^2 + z^2}$ is distance from the origin.

2. Why does cylindrical coordinates have Jacobian r? **Answer:** Cylindrical Jacobian is r because polar area elements scale by radius.

3. Why does spherical coordinates have Jacobian $\rho^2 \sin\phi$? **Answer:** Spherical Jacobian is $\rho^2 \sin\phi$ because shells have radial scale ρ^2 and polar-angle scale $\sin\phi$.

4. What surfaces are represented by $r = a$, $\theta = c$, and $z = c$? **Answer:** $r = a$ is a cylinder; $\theta = c$ is a vertical half-plane; $z = c$ is a horizontal plane.

5. What surfaces are represented by $\rho = a$, $\phi = c$, and $\theta = c$? **Answer:** $\rho = a$ is a sphere; $\phi = c$ is a cone; $\theta = c$ is a vertical half-plane.

6. Convert $x^2 + y^2 = 4$ into cylindrical coordinates. **Answer:** $x^2 + y^2 = 4$ becomes $r = 2$.

7. Convert $x^2 + y^2 + z^2 = 9$ into spherical coordinates. **Answer:** $x^2 + y^2 + z^2 = 9$ becomes $\rho = 3$.

8. Convert $z = \sqrt{x^2 + y^2}$ into spherical coordinates. **Answer:** $z = \sqrt{x^2 + y^2}$ becomes $\phi = \pi/4$.

9. In cylindrical coordinates, why does $z = 1 - x^2 - y^2$ become $z = 1 - r^2$? **Answer:** Since $x^2 + y^2 = r^2$, $z = 1 - x^2 - y^2$ becomes $z = 1 - r^2$.

10. If a region is inside a sphere and above a cone, which coordinates are usually best? **Answer:** Spherical coordinates.

11. If a region is inside a cylinder and below a plane, which coordinates are usually best? **Answer:** Cylindrical coordinates.

12. What does the inequality $0 \le \theta \le \pi/2$ usually mean geometrically? **Answer:** First quadrant in the xy-plane.

13. What does the inequality $0 \le \phi \le \pi/4$ mean geometrically? **Answer:** Region near the positive z-axis above the cone $\phi = \pi/4$.

14. How do you decide whether r or z should be integrated first? **Answer:** Choose the order that makes bounds simplest; usually integrate the variable with simple top-bottom bounds first.

15. What is the biggest setup mistake students make in cylindrical/spherical coordinates? **Answer:** Forgetting the Jacobian.

B. Change Of Variables

1. What is a Jacobian? **Answer:** A Jacobian is the determinant measuring area/volume scaling under a coordinate change.

2. Why do we take the absolute value of the Jacobian in change of variables? **Answer:** Absolute value is used because area/volume is positive.

3. If the Jacobian is constant, what does that say geometrically? **Answer:** Constant Jacobian means constant area-scaling.

4. What is the first step when using a given change of variables? **Answer:** Transform the boundary.
5. Why should boundary curves be transformed before the integrand? **Answer:** Boundaries determine the new integration limits.
6. How do you know the new uv-region is simpler than the old xy-region? **Answer:** It usually turns curved boundaries into straight lines or circles.
7. What happens if the Jacobian is zero? **Answer:** The coordinate map degenerates and is not locally invertible.
8. Why is polar coordinates just a special case of change of variables? **Answer:** Polar coordinates are $x = r\cos\theta$, $y = r\sin\theta$.
9. In $x = r\cos\theta$, $y = r\sin\theta$, what is the Jacobian? **Answer:** The polar Jacobian is r.
10. If $x = 2u$, $y = 3v$, what is the Jacobian and what does it mean? **Answer:** For $x = 2u$, $y = 3v$, the Jacobian is 6.

C. Vector Fields And Line Integrals

1. What is a vector field? **Answer:** A vector field assigns a vector to each point.
2. What does $\int_C \mathbf{F} \cdot d\mathbf{r}$ measure? **Answer:** $\int_C \mathbf{F} \cdot d\mathbf{r}$ measures work or circulation along a path.
3. Why does orientation matter for vector line integrals? **Answer:** Reversing orientation changes the sign.
4. Why does orientation not matter for scalar line integrals $\int_C f\, ds$? **Answer:** Scalar arc-length integrals use ds, which is always positive.
5. How do you parametrize a circle of radius a counterclockwise? **Answer:** $\mathbf{r}(t) = (a\cos t, a\sin t)$, $0 \le t \le 2\pi$.
6. What is $d\mathbf{r}$ in terms of a parametrization $\mathbf{r}(t)$? **Answer:** $d\mathbf{r} = \mathbf{r}'(t)\, dt$.
7. When is a line integral path independent? **Answer:** A line integral is path independent when the field is conservative.
8. What is a conservative vector field? **Answer:** Conservative means $\mathbf{F} = \nabla\phi$.
9. How do you test whether $\langle P, Q\rangle$ is conservative in $\mathbb{R}^2$? **Answer:** In $\mathbb{R}^2$, check $P_y = Q_x$ on a simply connected domain.
10. How do you find a potential function once you know the field is conservative? **Answer:** Integrate one component, then compare partial derivatives to find the missing function.
11. Why is the domain important when checking if a field is conservative? **Answer:** Domains with holes can break global conservativeness.
12. What is special about $\left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right\rangle$? **Answer:** It has zero curl away from the origin but nonzero circulation around the origin.
13. Why can a field have zero curl but still fail to be globally conservative? **Answer:** Because the domain may not be simply connected (e.g., it has a hole).

14. What does the Fundamental Theorem for Line Integrals say? **Answer:** If $\mathbf{F} = \nabla\phi$, then $\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$.

15. How is it similar to the one-variable Fundamental Theorem of Calculus? **Answer:** It is the multivariable version of $\int_a^b f'(x)\,dx = f(b) - f(a)$.

D. Green's Theorem, Curl, Divergence

1. State Green's theorem. **Answer:** Green's theorem: $\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA$.

2. What does "positively oriented" mean? **Answer:** Positive orientation means counterclockwise.

3. In Green's theorem, why does $Q_x - P_y$ appear? **Answer:** $Q_x - P_y$ is scalar curl.

4. What is the scalar curl of $\langle P, Q\rangle$? **Answer:** Scalar curl is $Q_x - P_y$.

5. How can Green's theorem simplify a hard boundary integral? **Answer:** It replaces a boundary integral by an area integral.

6. What is the curl of a 3D vector field? **Answer:** $\text{curl}\langle P, Q, R\rangle = \langle R_y - Q_z, P_z - R_x, Q_x - P_y\rangle$.

7. What does curl measure conceptually? **Answer:** Curl measures local rotation.

8. What is divergence? **Answer:** Divergence is $P_x + Q_y + R_z$.

9. What does divergence measure conceptually? **Answer:** Divergence measures source/sink behavior.

10. If $\mathbf{F} = \nabla\phi$, what is $\text{curl}\,\mathbf{F}$? **Answer:** If $\mathbf{F} = \nabla\phi$, then $\text{curl}\,\mathbf{F} = \mathbf{0}$.

11. If $\text{div}\,\mathbf{F} > 0$, what does that suggest physically? **Answer:** Positive divergence suggests outward flow/source.

12. How do you remember the formula for curl? **Answer:** Remember curl as $\nabla \times \mathbf{F}$.

13. What is the difference between curl and divergence? **Answer:** Curl measures rotation; divergence measures expansion/outflow.

14. What is the difference between Green's theorem and the Fundamental Theorem for Line Integrals? **Answer:** FTLI is for conservative fields and endpoints; Green's theorem is for closed curves and enclosed regions.

15. Why must Green's theorem use a closed curve? **Answer:** Green's theorem needs a boundary of a region.

4 Direct Practice Questions From Problem Sets

These are the most relevant questions to prepare for Quiz 5. They are copied or lightly standardized from Problem Sets 8, 9, and 10.

From Problem Set 8

1. PS8 Problem 7(i): Evaluate $\iiint_V y\,dV$, where $V = \{0 \le x \le 3,\ 0 \le y \le x,\ x - y \le z \le x + y\}$.
2. PS8 Problem 7(iii): Evaluate $\iiint_V \sin y\,dV$, where V is below $z = x$ and above the triangular region with vertices $(0,0,0)$, $(\pi,0,0)$, $(0,\pi,0)$.
3. PS8 Problem 7(iv): Evaluate $\iiint_V xy\,dV$, where V is bounded by $y = x^2$, $x = y^2$, $z = 0$, and $z = x+y$.
4. PS8 Problem 8(i): Change the order of $\int_0^1 \int_0^y f(x,y)\,dx\,dy$.
5. PS8 Problem 8(ii): Change the order of $\int_0^2 \int_{2y}^{y^2} f(x,y)\,dx\,dy$.
6. PS8 Problem 9(i): Write three other iterated integrals for the region $0 \le x \le y \le z \le 1$.

From Problem Set 9

1. PS9 Problem 1(i): Use $x = u^2 - v^2$, $y = 2uv$ to evaluate $\iint_R y\,dA$, where R is bounded by $y = 0$, $y^2 = 4 - 4x$, and $y^2 = 4 + 4x$ with $y \ge 0$.
2. PS9 Problem 1(ii): Use $x = 2u + v$, $y = u + 2v$ to evaluate $\iint_R (x - 3y)\,dA$, where R has vertices $(0,0)$, $(2,1)$, $(1,2)$.
3. PS9 Problem 2(ii): Evaluate $\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{2} (x^2 + y^2)\,dz\,dy\,dx$.
4. PS9 Problem 4(i): Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} xy\,dz\,dy\,dx$ using spherical coordinates.
5. PS9 Problem 6(i): Find the volume above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2+y^2+z^2 = z$.
6. PS9 Problem 7(ii): Determine whether $F = (e^x \sin y, e^x \cos y)$ is conservative. If yes, find ϕ.
7. PS9 Problem 8(v): Determine whether $F = (e^x \sin yz, ze^x \cos yz, ye^x \cos yz)$ is conservative. If yes, find ϕ.

From Problem Set 10

1. PS10 Problem 1(i): Show path independence and evaluate $\int_C 2xe^{-y}\,dx + (2y - x^2e^{-y})\,dy$ from $(1,0)$ to $(2,1)$.
2. PS10 Problem 1(ii): Show path independence and evaluate $\int_C \sin y\,dx + (x\cos y - \sin y)\,dy$ from $(2,0)$ to $(1,\pi)$.
3. PS10 Problem 2(i): Compute $\int_C \left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right\rangle \cdot d\mathbf{r}$ over the unit circle counterclockwise.
4. PS10 Problem 3(i): Use Green's theorem to evaluate $\int_C xy^2\,dx + 2x^2y\,dy$ over the triangle with vertices $(0,0)$, $(2,2)$, $(2,4)$.
5. PS10 Problem 3(ii): Use Green's theorem to evaluate $\int_C \cos y\,dx + x^2 \sin y\,dy$ over the rectangle with vertices $(0,0)$, $(5,0)$, $(5,2)$, $(0,2)$.
6. PS10 Problem 3(iii): Use Green's theorem to evaluate $\int_C x\,dy - y\,dx$ over the circle of radius a.

5 Short Answer Sheet

Theory Quiz Short Answers

1. $r = \sqrt{x^2 + y^2}$ is distance from the z-axis; $\rho = \sqrt{x^2 + y^2 + z^2}$ is distance from the origin.
2. Cylindrical Jacobian is r because polar area elements scale by radius.
3. Spherical Jacobian is $\rho^2 \sin\phi$ because shells have radial scale ρ^2 and polar-angle scale $\sin\phi$.
4. $r = a$ is a cylinder; $\theta = c$ is a vertical half-plane; $z = c$ is a horizontal plane.
5. $\rho = a$ is a sphere; $\phi = c$ is a cone; $\theta = c$ is a vertical half-plane.
6. $x^2 + y^2 = 4$ becomes $r = 2$.
7. $x^2 + y^2 + z^2 = 9$ becomes $\rho = 3$.
8. $z = \sqrt{x^2 + y^2}$ becomes $\phi = \pi/4$.
9. Since $x^2 + y^2 = r^2$, $z = 1 - x^2 - y^2$ becomes $z = 1 - r^2$.
10. Spherical coordinates.
11. Cylindrical coordinates.
12. First quadrant in the xy-plane.
13. Region near the positive z-axis above the cone $\phi = \pi/4$.
14. Choose the order that makes bounds simplest; usually integrate the variable with simple top-bottom bounds first.
15. Forgetting the Jacobian.
16. A Jacobian is the determinant measuring area/volume scaling under a coordinate change.
17. Absolute value is used because area/volume is positive.
18. Constant Jacobian means constant area-scaling.
19. Transform the boundary.
20. Boundaries determine the new integration limits.
21. It usually turns curved boundaries into straight lines or circles.
22. The coordinate map degenerates and is not locally invertible.
23. Polar coordinates are $x = r\cos\theta$, $y = r\sin\theta$.
24. The polar Jacobian is r.
25. For $x = 2u$, $y = 3v$, the Jacobian is 6.
26. A vector field assigns a vector to each point.
27. $\int_C \mathbf{F} \cdot d\mathbf{r}$ measures work or circulation along a path.
28. Reversing orientation changes the sign.

29. Scalar arc-length integrals use ds, which is always positive.

30. $\mathbf{r}(t) = (a\cos t, a\sin t)$, $0 \le t \le 2\pi$.

31. $d\mathbf{r} = \mathbf{r}'(t)\,dt$.

32. A line integral is path independent when the field is conservative.

33. Conservative means $\mathbf{F} = \nabla\phi$.

34. In $\mathbb{R}^2$, check $P_y = Q_x$ on a simply connected domain.

35. Integrate one component, then compare partial derivatives to find the missing function.

36. Domains with holes can break global conservativeness.

37. It has zero curl away from the origin but nonzero circulation around the origin.

38. If $\mathbf{F} = \nabla\phi$, then $\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(B) - \phi(A)$.

39. It is the multivariable version of $\int_a^b f'(x)\,dx = f(b) - f(a)$.

40. Green's theorem: $\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA$.

41. Positive orientation means counterclockwise.

42. $Q_x - P_y$ is scalar curl.

43. Scalar curl is $Q_x - P_y$.

44. It replaces a boundary integral by an area integral.

45. $\operatorname{curl}\langle P, Q, R\rangle = \langle R_y - Q_z, P_z - R_x, Q_x - P_y\rangle$.

46. Curl measures local rotation.

47. Divergence is $P_x + Q_y + R_z$.

48. Divergence measures source/sink behavior.

49. If $\mathbf{F} = \nabla\phi$, then $\operatorname{curl}\mathbf{F} = \mathbf{0}$.

50. Positive divergence suggests outward flow/source.

51. Remember curl as $\nabla \times \mathbf{F}$.

52. Curl measures rotation; divergence measures expansion/outflow.

53. FTLI is for conservative fields and endpoints; Green's theorem is for closed curves and enclosed regions.

54. Green's theorem needs a boundary of a region.

Selected Problem-Set Answers

Question	Short answer
PS8 7(i)	$\displaystyle\int_0^3\int_0^x\int_{x-y}^{x+y} y\,dz\,dy\,dx = \frac{27}{2}.$
PS8 7(iii)	$\displaystyle\frac{1}{2}\int_0^\pi (\pi-y)^2\sin y\,dy = \frac{\pi^2-4}{2}.$
PS8 7(iv)	Region $0\le x\le 1$, $x^2\le y\le\sqrt{x}$, $0\le z\le x+y$; value $\dfrac{3}{28}$.
PS8 8(i)	$\displaystyle\int_0^1\int_x^1 f(x,y)\,dy\,dx.$
PS8 8(ii)	Signed reversal: $\displaystyle\int_0^4\int_{\sqrt{x}}^{x/2} f(x,y)\,dy\,dx$. Geometric positive orientation: $-\int_0^4\int_{x/2}^{\sqrt{x}} f(x,y)\,dy\,dx$.
PS8 9(i)	Region $0\le x\le y\le z\le 1$. Examples: $\int_0^1\int_0^z\int_x^z f\,dy\,dx\,dz$, $\int_0^1\int_x^1\int_x^z f\,dy\,dz\,dx$, $\int_0^1\int_x^1\int_y^1 f\,dz\,dy\,dx$.
PS9 1(i)	New region $0\le u,v\le 1$, Jacobian $4(u^2+v^2)$, answer 2.
PS9 1(ii)	New region $u\ge 0$, $v\ge 0$, $u+v\le 1$, Jacobian 3, answer -3.
PS9 2(ii)	Cylindrical setup $\int_0^{2\pi}\int_0^2\int_r^2 r^3\,dz\,dr\,d\theta$, answer $\frac{16\pi}{5}$.
PS9 4(i)	Spherical setup $0\le\theta\le\frac{\pi}{2}$, $0\le\phi\le\frac{\pi}{4}$, $0\le\rho\le\sqrt{2}$; answer $\frac{4\sqrt{2}-5}{15}$.
PS9 6(i)	$0\le\rho\le\cos\phi$, $0\le\phi\le\frac{\pi}{4}$, $0\le\theta\le 2\pi$; volume $\frac{\pi}{8}$.
PS9 7(ii)	Conservative; potential $\phi = e^x\sin y + C$.
PS9 8(v)	Conservative; potential $\phi = e^x\sin(yz) + C$.
PS10 1(i)	Potential $\phi = x^2e^{-y} + y^2$; value $\frac{4}{e}$.
PS10 1(ii)	Potential $\phi = x\sin y + \cos y$; value -2.
PS10 2(i)	Parametrize unit circle; integral 2π.
PS10 3(i)	Green's theorem gives $\iint_D 2xy\,dA = \frac{56}{3}$.
PS10 3(ii)	Green's theorem gives $\int_0^5\int_0^2(2x+1)\sin y\,dy\,dx = 30(1-\cos 2)$.
PS10 3(iii)	Green's theorem gives $\iint_D 2\,dA = 2\pi a^2$.

6 Notation Table

Notation	Meaning
$\mathbf{F}$	Vector field.
P, Q, R	Component functions of a vector field.
$\nabla\phi$	Gradient of scalar potential ϕ.
$\operatorname{curl}\mathbf{F}$	Curl; measures local rotation.
$\operatorname{div}\mathbf{F}$	Divergence; measures source/sink behavior.
C	Curve, often boundary curve.
D	Plane region enclosed by C.
E or V	Solid region in $\mathbb{R}^3$.
dA	Area element.
dV	Volume element.
ds	Arc-length element.
$d\mathbf{r}$	Vector differential along a curve.
r	Cylindrical/polar radius $\sqrt{x^2+y^2}$.

ρ	Spherical radius $\sqrt{x^2+y^2+z^2}$.
θ	Angle in the xy-plane.
ϕ	Angle from positive z-axis in spherical coordinates.
J	Jacobian determinant.
$\oint_C$	Line integral around a closed curve.
$P_y = Q_x$	Two-dimensional conservative-field test on simply connected domains.
$Q_x - P_y$	Scalar curl; Green's theorem integrand.

7 FAQs

1. **When should I use cylindrical coordinates?** Use them when x^2+y^2 appears, or the region has a cylinder, disk, cone, or circular paraboloid.

2. **When should I use spherical coordinates?** Use them for spheres, balls, cones, or integrands involving $x^2+y^2+z^2$.

3. **What is the most common mistake in cylindrical coordinates?** Forgetting the factor r in $dV = r\,dz\,dr\,d\theta$.

4. **What is the most common mistake in spherical coordinates?** Forgetting $\rho^2 \sin\phi$ or mixing up ϕ and θ.

5. **Is ϕ measured from the xy-plane?** In Stewart convention, no. ϕ is measured from the positive z-axis.

6. **What does $z = \sqrt{x^2+y^2}$ mean geometrically?** A cone. In spherical coordinates it is $\phi = \pi/4$.

7. **How do I convert a sphere $x^2+y^2+z^2 = z$?** Use $\rho^2 = \rho\cos\phi$, so $\rho = \cos\phi$.

8. **Why does $x^2+y^2+z^2 = z$ describe a sphere?** Completing the square gives $x^2+y^2+(z-\frac{1}{2})^2 = \frac{1}{4}$.

9. **Why is there an absolute value around the Jacobian?** Orientation can flip, but area/volume must stay positive.

10. **How do I find a new region after change of variables?** Transform each boundary curve into equations in the new variables.

11. **Do I transform the integrand or the region first?** Usually transform the region first.

12. **What does a conservative field mean?** It is a gradient field $\mathbf{F} = \nabla\phi$.

13. **Why are conservative fields important?** Their line integrals depend only on endpoints.

14. **How do I find a potential in 2D?** Integrate P with respect to x, add $g(y)$, then compare with Q.

15. **How do I check if $\langle P, Q\rangle$ is conservative?** Check whether $P_y = Q_x$, assuming the domain is simply connected.

16. **What if $P_y = Q_x$ but the domain has a hole?** The field may fail to be globally conservative.

17. **What is the classic hole-domain example?** $\left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right\rangle$ on $\mathbb{R}^2 \setminus \{0\}$.

18. **What is the line integral of that field around a counterclockwise circle?** 2π, regardless of radius.

19. **Does zero curl always imply conservative?** Only on simply connected domains.

20. **What does Green's theorem do?** Converts a closed line integral into a double integral over the enclosed region.

21. **When should I use Green's theorem?** When the curve is closed and the boundary integral looks harder than the area integral.

22. **What is positive orientation?** Counterclockwise around the boundary.

23. **What happens if the curve is clockwise?** The answer changes sign.

24. **How do I remember Green's theorem?** $\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA$.

25. **What if $Q_x - P_y$ is constant?** Then the line integral is constant times the area.

26. **Why is $\int_C x\,dy - y\,dx = 2 \cdot \mathbf{Area}(D)$?** Here $P = -y$, $Q = x$, so $Q_x - P_y = 1 - (-1) = 2$.

27. **What is curl in 2D?** The scalar $Q_x - P_y$.

28. **What is curl in 3D?** $\langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$.

29. **What is divergence?** Sum of matching partial derivatives: $P_x + Q_y + R_z$.

30. **What is the difference between curl and divergence?** Curl measures rotation; divergence measures net outward flow.

31. **Do I need divergence theorem for this quiz?** The syllabus names divergence, but not necessarily the divergence theorem. Know the definition and interpretation.

32. **What is the fastest way to evaluate a conservative line integral?** Find ϕ, then compute $\phi(\text{end}) - \phi(\text{start})$.

33. **What if a line integral says "any path"?** It is hinting at path independence/conservative fields.

34. **How do I parametrize a circle radius a?** $\mathbf{r}(t) = (a\cos t, a\sin t)$, $0 \le t \le 2\pi$.

35. **How do I parametrize clockwise?** Use $\mathbf{r}(t) = (a\cos t, -a\sin t)$ or reverse the limits.

36. **What is the answer if the vector field is a gradient and the curve is closed?** Zero.

37. **Why?** The start and end points are the same, so $\phi(B) - \phi(A) = 0$.

38. **How do I decide bounds for a triple integral?** Draw/project the solid, choose outer variables for the projection, then set inner bounds between lower and upper surfaces.

39. **What is a good quiz strategy?** First identify the theorem/coordinate system, then write the correct setup before simplifying.

40. **What should I memorize?** Cylindrical formulas, spherical formulas, Jacobians, Green's theorem, FTLI, curl, divergence, and conservative-field tests.

41. **What should I practice most?** Setting up bounds and recognizing when a line integral is path independent.